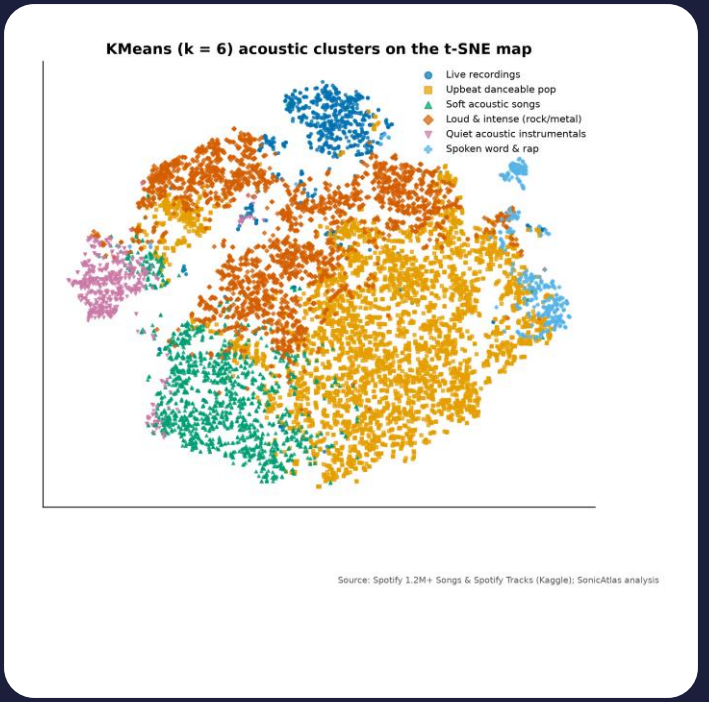
S O N I C A T L A S

An Interactive In-Browser EDA Platform for Spotify Audio Features and Lyrics Across Genres and Eras

Final Review | Complete project

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Real acoustic map of 10,000 tracks (t-SNE)

BACSE301 Exploratory Data Analysis (Project-based Course)
Faculty: _____
School of Computer Science and Engineering, VIT Vellore | October 2026

ABSTRACT

Mapping how the sound of music changes across genres and eras

Spotify's archived audio descriptors (energy, valence, danceability, acousticness and more) are rarely explored beyond static charts. SonicAtlas cleans 1,204,025 tracks released 1900-2020, a genre table of 89,740 tracks in 114 genres and 161,401 English lyrics, explores them with the course methods (descriptive statistics, correlation, PCA/t-SNE/LLE, time series, hypothesis tests, text mining), and publishes everything in a website that runs SQL inside the browser with DuckDB-WASM.

1,204,025

tracks analysed

114

genres compared

1959-2020

years of trends

Prior work is static and studies sound and words apart

Study	What they did	Gap SonicAtlas addresses
J. Serrà et al. (2012)	Tracked pitch, timbre and loudness of Western pop over 50+ years	Static figures; no genre-level or interactive view
M. Interiano et al. (2018)	Linked audio features of 500k UK releases to chart success	No lyrics; results cannot be re-sliced by readers
C. O. Brand et al. (2019)	Measured rising negativity in song lyrics over 50 years	Lyrics only; audio mood not compared
L. van der Maaten et al. (2008)	Introduced t-SNE for 2-D maps of high-dimensional data	Global distances unreliable; no music benchmark
D. H. da Silva et al. (2025)	Streamlit web app for EDA and ML on medical data	Needs a server; not music-aware

Static results

fixed figures, no re-slicing

Sound vs words

audio and lyrics never compared

Data quality

zero tempos, re-issues ignored

Generic tools

not music-aware, need a server

25 verified papers reviewed in the report (Table I).

Open music data exists, but it is messy and locked in static charts

Messy data

2,777 zero-BPM tempos, 71,694 duplicate title+artist rows, no shared keys

Frozen results

papers and notebooks show fixed figures; readers cannot ask new questions

Split modalities

audio features and lyrics are analysed in isolation

Hosting cost

interactive tools usually need a running server

Objectives

1. Clean 1M+ tracks; validate MICE by masking 5% of values
2. Descriptive stats in course conventions (n-1, Tukey IQR, G1, G2)
3. Correlation and multivariate genre fingerprints
4. Compare PCA, t-SNE, LLE, UMAP; name acoustic clusters
5. Trends, STL seasonality, change points, forecast
6. ANOVA, chi-square, feature selection; lyrics sentiment
7. In-browser SQL with lite-data queries under 1 second

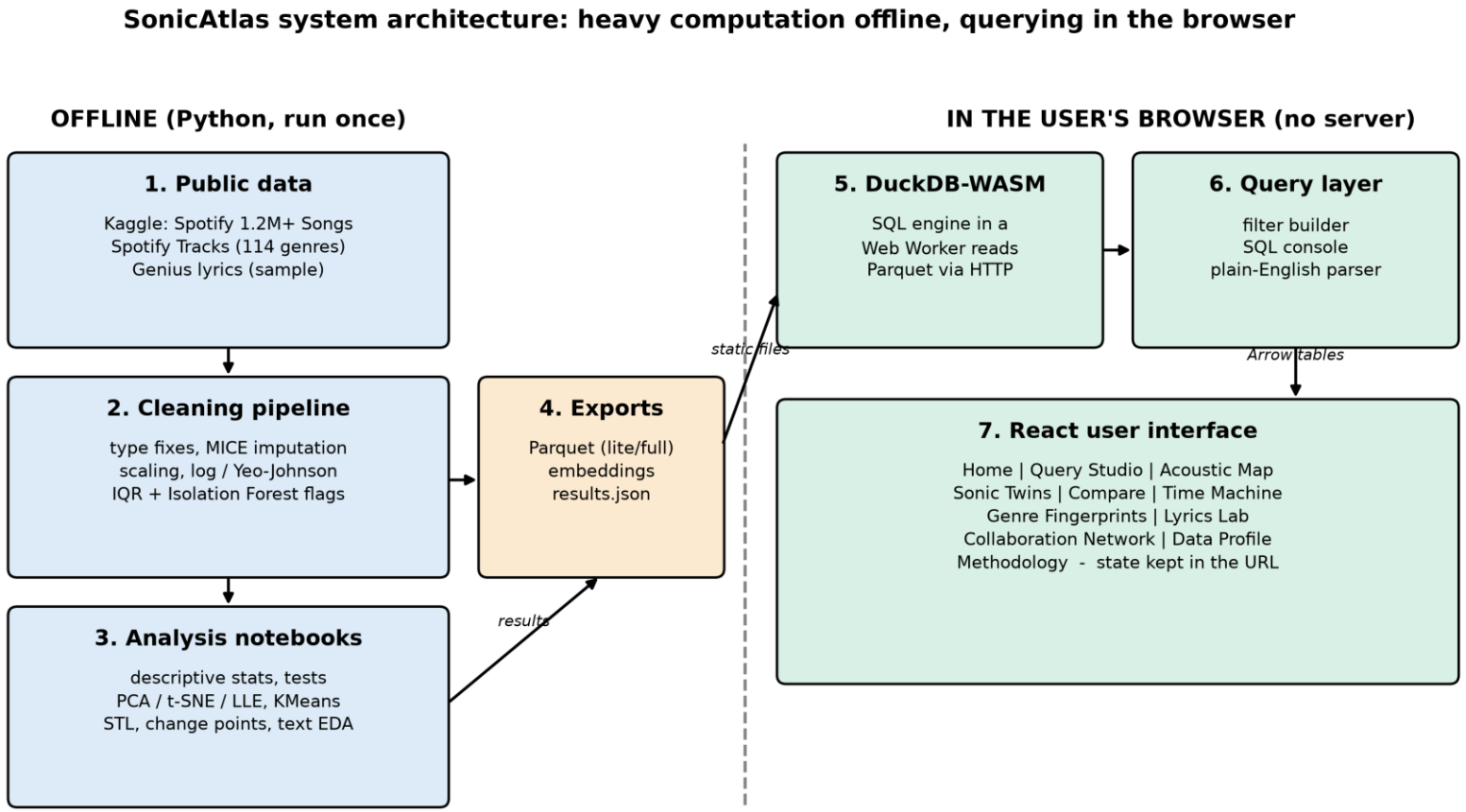
EXISTING SYSTEMS VS PROPOSED SYSTEM

SonicAtlas combines what existing tools offer separately

System	Open data	Music-specific EDA	Audio + lyrics	Interactive	Serverless
Spotify app / Web API	No (closed)	No	No	Yes (app)	No
Static notebooks / papers	Yes	Partly	Rarely	No	No
Streamlit EDA app [25]	Yes	Generic	No	Yes	No (server)
YData / Sweetviz / D-Tale	Yes	Generic	No	Partly	No (server for D-Tale)
SonicAtlas (ours)	Yes	Yes	Yes	Yes	Yes

Only SonicAtlas is open, music-aware, combines sound with words, and runs entirely in the browser.

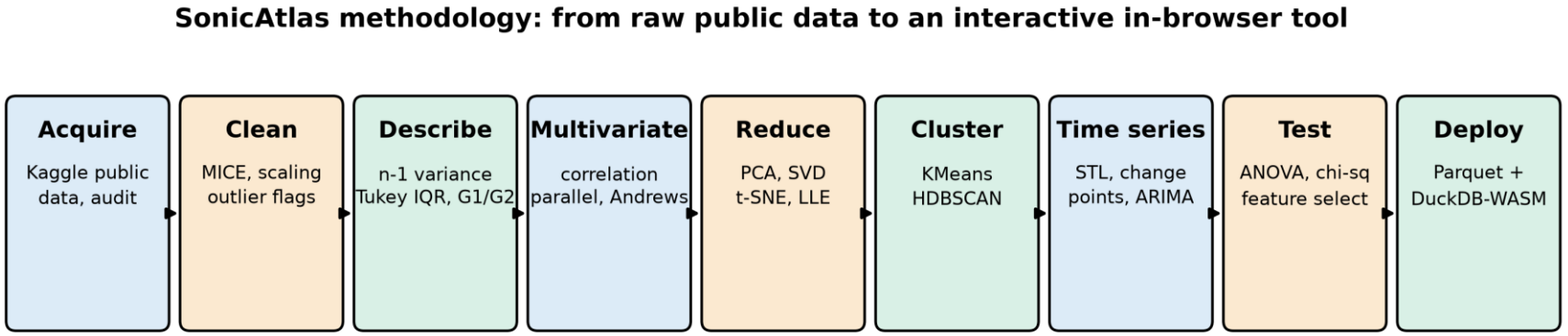
Heavy maths runs offline; querying runs in the user's browser



SonicAtlas design (Team 05)

METHODOLOGY

From raw public files to an interactive in-browser tool



SonicAtlas design (Team 05)

- Course conventions: n-1 variance, Tukey quartiles, adjusted G1, bias-corrected G2
- Validated MICE (energy error -58% vs mean); outliers flagged, never dropped
- Embeddings on a 10,000-track genre-stratified sample; trends on canonical releases only
- Every number in documents comes from results.json (cross-checked by script)

KEY FINDINGS

Five findings, all measured

+7.8 dB

louder music 1959-2020

r = 0.82

loudness vs energy

42%

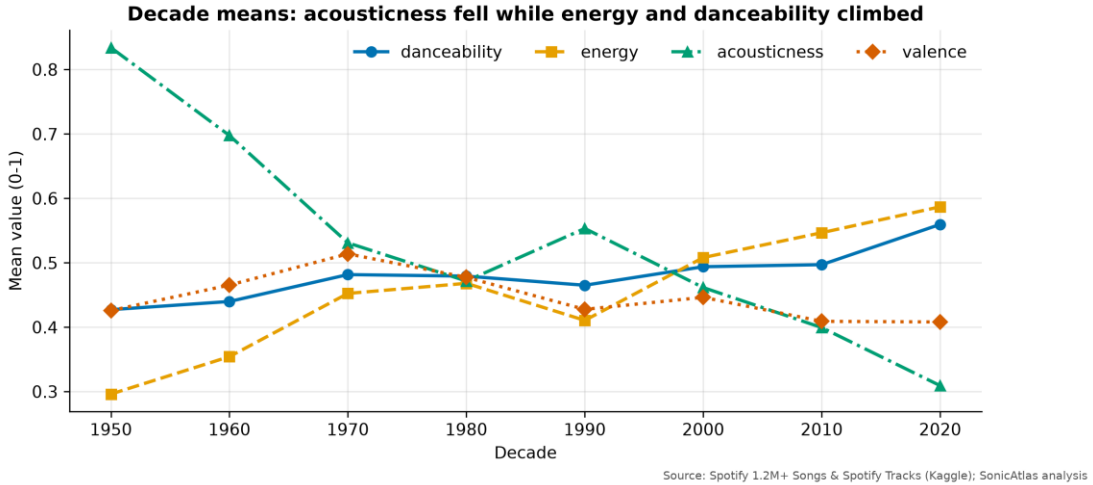
of danceability explained by genre

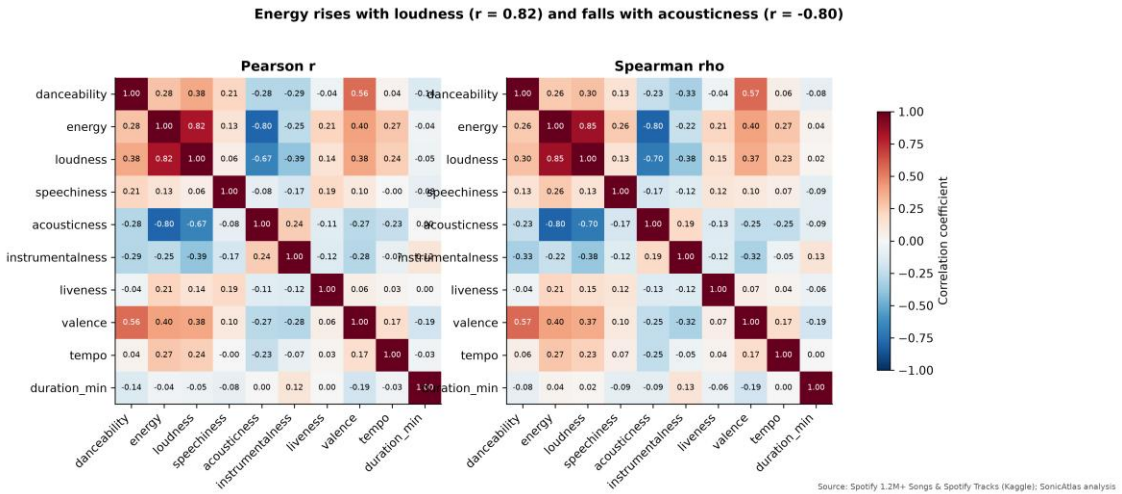
0.976

t-SNE trustworthiness (best)

r = 0.06

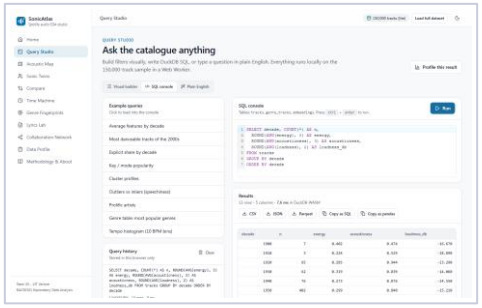
lyric sentiment vs audio valence





LIVE DEMO AND OUTCOME

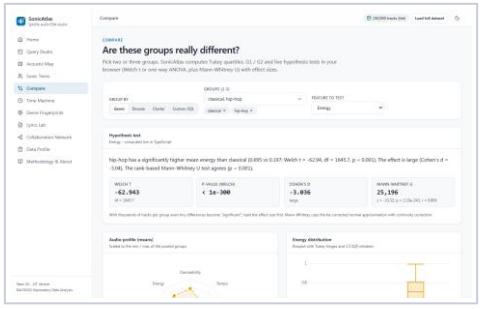
SonicAtlas live, plus an IEEE paper prepared for submission



Query Studio



Acoustic Map



Compare



SonicAtlas: An Interactive In-Browser Exploratory Data Analysis Platform for Multi-Dimensional Audio Features and Lyrics of Spotify Tracks Across Genres and Eras

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***Abstract*—Archived Spotify audio descriptors are among the few open windows onto the acoustic character of recorded music, but most analyses of them are static and stop at a handful of charts. We describe SonicAtlas, a reproducible exploratory pipeline and a serverless web platform built on 1,204,025 tracks, 89,740 genre-labelled tracks in 114 genres and a 161,401-song English lyrics sample. Impossible values are repaired with chained-equation imputation that lowers masked-value error by up to 57.5% relative to mean filling, and outliers are flagged rather than removed. Across 1959-2020, average loudness climbs by 7.8 dB while acousticness falls from 0.86 to 0.31, with loudness change points at 1989, 1999. Genre accounts for 42.5% of the variance of danceability, yet genre labels overlap strongly in every two-dimensional embedding; t-SNE attains the highest trustworthiness (0.976) among PCA, LLE, UMAP and t-SNE. Lyric sentiment is almost unrelated to audio valence ($r \approx 0.06$). All results are explorable in the browser through SQL executed by DuckDB-WASM on Parquet files, with no server.**

***Index Terms*—exploratory data analysis, music information retrieval, audio features, dimensionality reduction, time-series analysis, in-browser analytics**

I. Introduction

Popular music has been studied quantitatively for more than a decade, from the long-run homogenisation of timbre and loudness [1] to revolutions in US chart music [2] and the predictability of commercial success [3], [4]. Streaming platforms made such work easier by publishing per-track descriptors such as energy, valence and danceability, and large crawls of these values still circulate on open-data portals even though the original API endpoint has been closed to new applications.

Two practical problems limit the value of these crawls. First, the raw tables contain impossible values, duplicated re-issues and sources that do not share keys, and published studies rarely say how such issues were treated. Second, results are usually frozen into figures, so a reader cannot ask a slightly different question without rebuilding the pipeline. This paper addresses both problems. Our contributions are: (i) a documented cleaning pipeline over 1.2 million tracks with validated imputation and explicit outlier and re-issue flags; (ii) a systematic exploration that combines course-standard descriptive statistics, multivariate views, four embedding methods, clustering, time-series decomposition, hypothesis tests and lyrics mining; and (iii) SonicAtlas, a static web application in which every table can be filtered, queried and compared in the reader's own browser.

II. Related Work

Audio descriptors and genre. Large open collections such as the Million Song Dataset [5] and handcrafted features for genre recognition [6] established the use of numeric audio summaries, and deep models now learn such features directly from audio [7].

Trends and popularity. Studies of musical evolution [1], [2] and hit prediction [3], [4] report strong temporal signals but publish them as static results. Lyrics research shows rising negativity over time [8], usually without the matching audio view; sentiment is often scored with VADER [9]. Recommendation surveys [10] motivate similarity search over descriptors.

Embeddings and EDA tooling. PCA [11], LLE [12], t-SNE [13] and UMAP [14] are standard for maps of high-dimensional data. Automated EDA has been surveyed by Milo and Somech [15], and interactive charts are commonly specified with declarative grammars such as Vega-Lite [16]. On the systems side, embeddable analytical engines [17], their WebAssembly port [18] and scalable linked views [19] make serverless analysis feasible. The closest system in spirit is a Streamlit application for exploratory analysis of clinical data [20], which, unlike SonicAtlas, requires a Python server and is not domain-aware for music.

III. Datasets and Preprocessing

We combine three public Kaggle datasets: a crawl of 1,204,025 tracks with audio descriptors and release dates [21],

- IEEE-format paper, 6 pages: prepared for submission
- Targets: IEEE CONECCT 2027, ISMIR, IEEE VIS
- GitHub: [GitHub Link]
- Live: [Live Website Link]

Louder, less acoustic, and more than genre alone

Conclusion

Careful cleaning matters; music got louder and less acoustic; genre explains 42% of danceability but overlaps in every map; lyrics and sound carry different moods.

Limitations

Spotify features are model-derived; catalogue favours recent years; genre join only 10.7%; English-only lyrics; the 2020s have one year.

Future work

Open audio features, multilingual lyrics, user study.

[9] J. Serrà et al. (2012), Measuring the Evolution of Contemporary Western Popular Music
[11] C. Hutto et al. (2014), VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text
[16] L. van der Maaten et al. (2008), Visualizing Data using t-SNE
[25] D. H. da Silva et al. (2025), A Web Application for exploratory data analysis and classification of Parkinson's Disease patients using machine learning models on different datasets